

Sumit Kumar

PhD Scholar,
Computer Science & Engg,
Indraprastha Institute of Information Technology Delhi

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Summary

My research primarily focuses on the challenges and opportunities in networking for AI/ML systems. More specifically, my future work aims to design scalable, low-latency, multi-tenant network and system architectures for AI training and inference workloads.

Education

- Indraprastha Institute of Information Technology, Delhi** Delhi, India
Ph.D. Computer Science and Engineering (CSE) *Since Oct 2023*
 - Advisor: Dr. Rinku Shah
 - Course work: Programmable Networks, Distributed Systems, Object Oriented Programming: and Design, Graduate Systems and Research Methods.
 - GPA: 8.64 / 10
- Jawaharlal Nehru University, New Delhi** Delhi, India
M.Tech Statistical Computing (Data Communication) *2020 - 2022*
 - Thesis Title: Minimizing Energy Of Multi-Hop Routing Protocol In Wireless Sensor Network
 - Advisor: Dr. Karan Singh
 - GPA: 8.0 / 9
- Central University of South Bihar, Gaya** Bihar, India
M.Sc. in Computer Science *2018 - 2020*
 - Thesis Title: IPSec VPN (Implementation of IPSec Tunnel Between Two Sites)
 - GPA: 8.54 / 10
- Magadh University (Gaya College), Gaya** Bihar, India
Bachelor in Computer Application *2014 - 2017*
 - Score: 65 / 100

Professional Experience

- Teaching Assistant** New Delhi, India
Indraprastha Institute of Information Technology Delhi *2024 - 2025*
 - Conducted tutorials, assignment demos, quiz preparation, paper checking, and grading:
 - * Computer Networks (Monsoon 2024)
 - * Systems for AI (Monsoon 2025)
- Machine Learning Intern** Gurugram, India
CNH Industrial *Mar 2023 - Dec 2022*
 - Solve customer-centric problems in the agricultural machinery industry through NLP/generative AI.

Selected Publications

Accepted Papers

1. **Sumit Kumar**, Arjun, Naman, Ramanjeet, Meet, Praveen, Satananda, Abed, and Rinku, “**Simulating LLM training workloads for heterogeneous compute and network infrastructure**,” ACM SIGCOMM NAIC 2025. DOI: <https://dl.acm.org/doi/10.1145/3748273.3749212>.

Skills

- **Programming Languages:** C, C++, Python, Bash shell
- **Libraries/Software Packages:** torch, numpy, pandas, sklearn, matplotlib, PyTorch, NS-3, htsim
- **Software Tools:** Git, Docker, Visual Studio Code, Eclipse, Kubernetes
- **Miscellaneous:** Algorithms, Network, Data Structures, Problem Solving

Current Projects

- **Simulating LLM training workloads for heterogeneous compute and network infrastructure:** This work develops a heterogeneity-aware simulation framework for distributed LLM training that models GPU clusters with diverse compute and network capabilities. The simulator enables accurate training time prediction through support for non-uniform workload partitioning, custom device group configurations, flexible device-to-parallelism mappings (tensor, data, and pipeline parallel), and heterogeneous collective communication primitives. It bridges the gap between idealized homogeneous assumptions in existing tools and realistic, multi-GPU-generation cloud deployments and model-specific heterogeneous deployment.
- **Fast simulation of Scale-Up Fabric and RDMA NIC for Heterogeneous Training:** Leveraging packet-level network simulation htsim, this approach models realistic intra-node and inter-node data paths in heterogeneous LLM training environments. The simulator emulates detailed flow-level behaviours of scale-up interconnects (e.g., NVLink, NVSwitch, UALink) and RDMA-enabled NICs, accurately reflecting congestion, queuing, and bandwidth sharing across multiple GPU generations and network hierarchies. It enables evaluation of end-to-end collective communication performance, distributed job scheduling, and resource contention in mixed hardware clusters, facilitating system optimization for large-scale, heterogeneous deployments.
- **In-Network Computing (INC) Enabled Switch Modeling for Collective Communication in ns-3:** Leveraging flow-level network simulation, this project extends the ns-3 switch model to support in-network computing for accelerating collective communication in distributed training workloads. The enhanced switch performs on-the-fly aggregation operations (e.g., reduction for AllReduce), reducing network traffic, latency, and congestion compared to traditional host-based approaches. The simulator captures realistic datacenter behaviors including queuing, bandwidth sharing, and topology-aware communication across leaf-spine architectures. It enables evaluation of flow completion time, scalability, and efficiency of collective operations, facilitating the design and optimization of high-performance distributed training systems.
- **Ultra Ethernet Transport (UET) Simulation and Modeling for AI/HPC Workloads:** Leveraging detailed network simulation, this project models the Ultra Ethernet Transport (UET) stack to capture end-to-end communication semantics for large-scale AI and HPC workloads. The simulator incorporates key UET components including the Semantic Sublayer (SES), Packet Delivery Sublayer (PDS), and Congestion Management Sublayer (CMS), enabling realistic modeling of send/receive, RDMA, and collective communication patterns. It captures critical transport behaviors such as reliable/unreliable delivery, ordering guarantees, congestion control, and flow control across leaf-spine datacenter topologies. The framework enables evaluation of latency, throughput, scalability, and network efficiency, facilitating design space exploration and optimization of next-generation high-performance Ethernet-based distributed systems.

Awards and Fellowship

- **Letter of Appreciation:** Volunteered and conducted a hands-on session on Linux networking at the ACM Winter School 2023.
- **Master of Technology Fellowship:** Awarded Non-NET Postgraduate Scholarship for a 2-year master's program.
- **NET:** Qualified the National Eligibility Test in Computer Science in 2023.

Academic Service

- Presented work-in-progress paper and poster at the IndoSys 2026 conference.
- Volunteered for the IndoSys 2026 conference.
- Served as a Program Committee member for ACM CAIS'26 Artifact Evaluation
- Served as a Program Committee member for MLSys 2026 Artifact Evaluation, assessing the correctness, usability, and reproducibility of systems and ML research artifacts.
- Attended the Google India Networking Research Summit 2026
- Presented a poster on my research work at the Research Innovation and Incubation Showcase Events (RIISE) 2025.
- Attended the SIGCOMM 2025 conference and presented short paper at SIGCOMM NAIC workshop in Combra, Portugal.
- Presented a poster at CSE Day 2025 IIITD.
- Attended the ICDCN Conference held at IIT Hyderabad.
- Volunteered and conducted a hands-on session on Linux networking at the ACM Winter School 2023.

Membership and Activities

- Student Member, Association for Computing Machinery (ACM), Member ID: 2264601, since July 2025.

References

- **Rinku Shah**
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Google Scholar: <https://scholar.google.co.in/citations?user=jNhSvOEAAAAJ>